From Reaction to Imagination: A Survey on World Models

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August 4, 2026

Abstract

This survey reconstructs the taxonomy of world models around two necessary criteria: **interactivity** and **predictive imagination**. Interactivity means that the model is coupled to actions, controls, or interventions in an environment loop. Predictive imagination means that the model can represent counterfactual futures, rather than only recognize the present or emit the next action. Under this definition, many vision-language models and many vision-language-action policies are neighboring categories rather than world models: VLMs are passive semantic encoders, while reactive VLAs map current observations to actions without an exposed future-state prediction interface. Agent-facing world models are organized by primary interface into Latent WM, Vision WM, and WAM. We instantiate a common probabilistic controlled state-space view, trace the evolution from recurrent latent imagination to video/BEV generation and joint world-action diffusion, and analyze an empirical trade-off among geometry/contact awareness, rollout latency, and token/sampling budget. The survey further specifies a hybrid neural-proposer/executable-verifier protocol with typed rejection and repair, proposes a layered open-loop, closed-loop, causal, safety, and efficiency evaluation protocol, and grounds the analysis in autonomous-driving and tactile–visual manipulation case studies. Across these routes, reliable planning depends on physical fidelity, uncertainty-aware verification, and explicit latency and compute budgets. The resulting view treats world modeling as a resource-constrained causal decision system rather than a synonym for video generation or multimodal policy learning.

Index Terms—World model, embodied AI, predictive imagination, Vision-Language-Action, latent dynamics, video simulation, WAM, hybrid physics-neural model.

1 Introduction and Taxonomy Shift

The phrase "world model" has become overloaded. It is used for latent dynamics models in reinforcement learning, large video generators, driving simulators, VLM-based agents, VLA robot policies, and even text-only reasoning systems. This v1 survey adopts a stricter operational definition: a world model for an agent must satisfy both **interactivity** and **predictive imagination**.

Interactivity means that model inputs or outputs participate in an action loop. A world model must condition on actions, controls, trajectories, goals, or interventions, or it must produce states that are usable for planning, control, verification, or simulation. **Predictive imagination** means that the model can represent and compare counterfactual futures. A minimal transition form is

$$\hat{S}_{t+k} = f_\theta(S_t, A_{t:t+k-1}, C), \tag{1}$$

where S_t is a latent, visual, geometric, or hybrid state; A denotes action or control; and C denotes context such as language, map, embodiment, or task constraints. The model is not merely labeling the current world. It is estimating how the world would change if an intervention were taken.

Definition. An *agent-facing world model* is a learned predictive system that (i) represents an environment state or belief from interaction history, (ii) defines an action- or intervention-conditioned distribution over one or more possible future states, and (iii) exposes those futures to planning, control, evaluation, or verification. We apply a paired-action interface test: hold history h_t fixed, query two feasible actions $a^{(1)} \neq a^{(2)}$, and require inspectable predictive outputs $F(h_t, a^{(1)})$ and $F(h_t, a^{(2)})$ whose divergence can be evaluated against the actions' observed consequences. A temporal encoder may contain predictive features internally, but a policy is classified as reactive when it cannot expose or score such action-conditioned futures for counterfactual comparison. A model need not reconstruct pixels, but its predictive state must preserve variables sufficient to distinguish action-relevant futures.

1.1 Probabilistic and State-Space Definitions

The deterministic expression above hides three properties that matter in practice: partial observability, stochastic futures, and epistemic uncertainty. A probabilistic world model instead defines a distribution over future states, observations, rewards, and termination variables,

$$p_\theta(s_{t+1:t+H}, o_{t+1:t+H}, r_{t:t+H}, d_{t:t+H} \mid h_t, a_{t:t+H-1}, c), \tag{2}$$

where $h_t = (o_{\le t}, a_{<t})$ is the observable history. This form separates aleatoric uncertainty, such as multiple physically plausible futures, from epistemic uncertainty caused by insufficient data or out-of-distribution intervention. A useful agent-facing model should expose both because a multimodal but well-supported future and an unsupported hallucination require different planning responses.

Most learned world models can also be written as a controlled state-space model:

$$z_t \sim q_\phi(z_t \mid z_{t-1}, a_{t-1}, o_t), \tag{3}$$

$$z_{t+1} \sim p_\theta(z_{t+1} \mid z_t, a_t, c), \tag{4}$$

$$o_t \sim p_\theta(o_t \mid z_t), \qquad r_t \sim p_\theta(r_t \mid z_t, a_t), \tag{5}$$

where q_ϕ filters observations into a posterior belief, $p_\theta(z_{t+1} \mid z_t, a_t, c)$ is the learned controlled transition, and the emission model reconstructs pixels, features, geometry, reward, or task progress. Latent WM emphasizes Eqs. 3–4; Vision WM invests more heavily in the emission or direct visual transition; WAM augments the state-space process with a joint action generator. This common SSM view makes their architectural differences comparable without assuming that the environment is fully observed.

1.2 MDP Transitions versus Counterfactual Futures

An MDP transition kernel $P(S_{t+1} \mid S_t, A_t)$ is a property of the environment when S_t is Markov. A learned world model only approximates that kernel from trajectories sampled under a behavior policy $\pi_{\mathcal{D}}$, so counterfactual use requires off-policy generalization to actions that may be rare or absent in the log:

$$p(S_{t+1:t+H} \mid h_t, \mathrm{do}(A_{t:t+H-1} = \bar{a}), C = c). \tag{6}$$

The practical gap is distributional rather than a claim that a learned model is a causal oracle. Logged data cover the state–action support induced by $\pi_{\mathcal{D}}$; a visual agent also observes only a belief over hidden mass, friction, intent, contact, and occluded objects. Long optimized action sequences can therefore leave training support and compound error. Counterfactual comparison evaluates mutually exclusive actions from the same factual history, but its credibility must be tested with held-out interventions or executable environments. A predictor trained only on behavioral correlations can confuse actions that co-occur with success with actions that cause success.

We therefore use *predictive imagination* to mean more than next-frame likelihood. A model should support interventions, retain uncertainty over hidden state, compare alternative futures from a shared history, and expose whether a proposed rollout lies outside its evidential support. This requirement is what separates causal planning utility from passive sequence completion.

This definition deliberately separates world models from two neighboring categories. Vision-language models such as CLIP, PaLM-E, and GPT-4V-style multimodal systems can encode or describe visual scenes, but they do not by themselves form an action-conditioned transition model [249, 73]. Vision-language-action policies such as RT-1, RT-2, OpenVLA, and Octo connect perception to action, but many are reactive policies: they map S_t or (o_t, ℓ) to A_t without explicit prediction of $\hat{S}_{t+1}$ under alternative actions [33, 32, 154, 292]. They are indispensable components, but not all of them are world models under the two-criterion definition.

Category	Interactive?	Predictive Imagination?	Typical Mapping	Status in This Survey
VLM	No	No	$(I, \ell) \rightarrow$ description, answer, embedding	Passive semantic encoder
VLA	Yes	Partial / interface-dependent	$(o_t, \ell, h_t) \rightarrow a_t$	Temporal features may be implicit; world-model status requires exposed counterfactual futures
Latent WM	Yes	Yes	$(z_t, a_t) \rightarrow z_{t+1}$	Compact predictive engine
Vision WM	Yes	Yes	$(I_t, A_t) \rightarrow \hat{I}_{t+1:t+k}$	Visual or geometric simulator
WAM	Yes	Yes	$P(V_{t+1:t+k}, A_{t:t+k} \mid H)$	Joint visual-action generator

Table 1: The v1 taxonomy uses two necessary criteria: interactivity and predictive imagination. The five visible categories are VLM, VLA, Latent WM, Vision WM, and WAM.

The shift matters because it changes the paper's spine. Instead of asking whether a system contains a vision model, transformer, diffusion module, or robot policy head, we ask whether it can imagine action-conditioned futures that matter for acting. This yields three agent-facing WM routes: **Latent WM**, **Vision WM**, and **WAM**. The rest of the survey follows that progression.

Contributions and organization. Section 2 diagnoses the objective-level limitations of VLM and reactive VLA interfaces. Section 3 develops the Latent WM, Vision WM, and WAM taxonomy and its algorithmic evolution. Section 4 preserves the time-dependent technology and lifecycle evidence. Section 5 formulates the geometry–latency–budget frontier and resource-aware optimization. Section 6 turns the hybrid proposal into an executable API with typed failure feedback. Section 7 aligns metrics and benchmarks with causal, closed-loop, safety, and compute requirements. Section 8 closes with driving and tactile–visual manipulation case studies plus shared open challenges.

2 Neighboring Interfaces: Perception and Reaction

2.1 VLM

VLMs are powerful perception and grounding systems. Contrastive encoders, large multimodal transformers, and image-language assistants can align objects, relations, scenes, instructions, and high-level commonsense. In embodied settings, they can identify a mug, parse a human command, localize a goal object, or provide semantic features for a downstream controller. This role is valuable, but it is not sufficient for world modeling.

The missing ingredient is temporal and interventional dynamics. CLIP-style contrastive training aligns an image with text but does not require the representation to preserve velocity, acceleration, object permanence, or action consequences [249]. PaLM-E-style multimodal language modeling improves embodied semantics and transfers visual observations into a language-model backbone, but token prediction alone does not identify a controlled physical transition [73]. A model can answer what is in an image or what a video depicts while remaining indifferent to which candidate intervention caused the change.

This objective mismatch explains why hidden physical variables are fragile. Two images can be semantically identical while corresponding to different masses, friction coefficients, support forces, joint limits, or contact modes. If contrastive or captioning loss does not require those variables, the encoder is free to collapse them. A cup can be correctly recognized as fragile without representing whether a push will slide it, rotate it around a contact point, topple it, or exceed the table boundary. Temporal pretraining helps only when its target preserves intervention-relevant state; ordering frames or predicting masked features is not automatically equivalent to learning causal mechanics.

Thus, VLMs are best interpreted as **passive semantic encoders**. They provide vocabulary and visual grounding for agents, but the predictive-imagination criterion is absent unless the VLM is embedded inside a larger action-conditioned transition system.

2.2 VLA

VLA systems add the missing action interface. RT-1, RT-2, OpenVLA, Octo, and related robot foundation policies map visual observations and language instructions to robot actions [33, 32, 154, 292]. They are interactive in the sense that their outputs affect the environment. However, many of them remain reactive: their dominant inference pattern is

$$\pi_\theta(a_t \mid o_t, \ell, h_t), \tag{7}$$

not

$$p_\theta(s_{t+1:t+H} \mid h_t, a_{t:t+H}, \ell, g). \tag{8}$$

The resulting limitation is **reactive blindness**. A policy can select a plausible next action while failing to compare future consequences. It may grasp the visible side of an object but fail to anticipate occlusion, collision, slippage, or irreversible state changes. It may obey the instruction but not ask whether an alternative action leads to a safer future. This is why this survey treats reactive VLA as a neighboring action interface, not as the core world-model category.

Reactive blindness has a concrete mechanism. During imitation or action-token training, the policy receives gradients for matching demonstrated actions, not for explaining their consequences. Under occlusion, the current observation can alias several hidden states: an object may be absent, temporarily hidden, or already displaced. A policy without an explicit belief update may issue the same action in all three states. Under irreversible transitions–dropping a fragile object, closing a latch, entering an intersection, or moving an obstacle into a dead end–a locally plausible action cannot be undone after the hidden consequence becomes visible. Long action chunks amplify the problem because later actions are committed before updated observations can correct the initial mistake. Safety-gap, trustworthiness, and execution-time verification benchmarks provide complementary empirical probes of these consequence-blind failures [80, 172, 204].

Table 2 separates these limitations from world-model capabilities. The relevant test is not whether a model internally contains temporal features, but whether its training and inference interface must predict action-conditioned futures and use them to alter decisions.

Interface	Training Pressure	Hidden Failure	Upgrade Needed
CLIP-like VLM	Align static image and text embeddings	Semantically similar states collapse despite different mass, contact, or support	Controlled temporal data and transition supervision
PaLM-E-like VLM	Predict language tokens from multimodal context	Embodied semantics without an identified action-conditioned dynamics law	Explicit belief state, future-state head, intervention data
RT-1/RT-2	Match demonstrated action tokens	Occluded states alias; action chunk follows correlation rather than simulated consequence	Candidate-action rollout and observation feedback
OpenVLA/Octo	Scale cross-embodiment policy learning	Distribution shift and irreversible actions expose missing counterfactual comparison	Uncertainty-aware world-model augmentation and verifier gating

Table 2: Why strong VLM and reactive VLA interfaces still fall short of the predictive-imagination criterion.

The distinction is not meant to diminish VLA. Rather, it identifies the next technical step: adding explicit future rollout, verifier-gated action selection, or joint visual-action generation so that VLA policies acquire predictive imagination.

3 True World Models: Interactive Predictive Engines

Figure 1 opens the core taxonomy. The first row contains VLM and VLA. The second row treats world modeling through Latent WM, Vision WM, and WAM. The third row compares their agent-facing capabilities: all three support reaction; Latent WM imagines implicitly in compressed state; Vision WM exposes imagination explicitly as visible or geometric futures; and WAM adds explicit action generation or scoring inside that imagined future.

These categories are not mutually exclusive architectures. They classify a system by its primary agent-facing predictive interface and optimization target. Dreamer-style models may decode pixels while remaining latent-first; a visual WAM spans explicit visual prediction and joint action generation; an omnimodal foundation model may expose several modes. Hybrid systems should therefore receive multiple mechanism tags even when one primary interface determines their placement in the comparison tables.

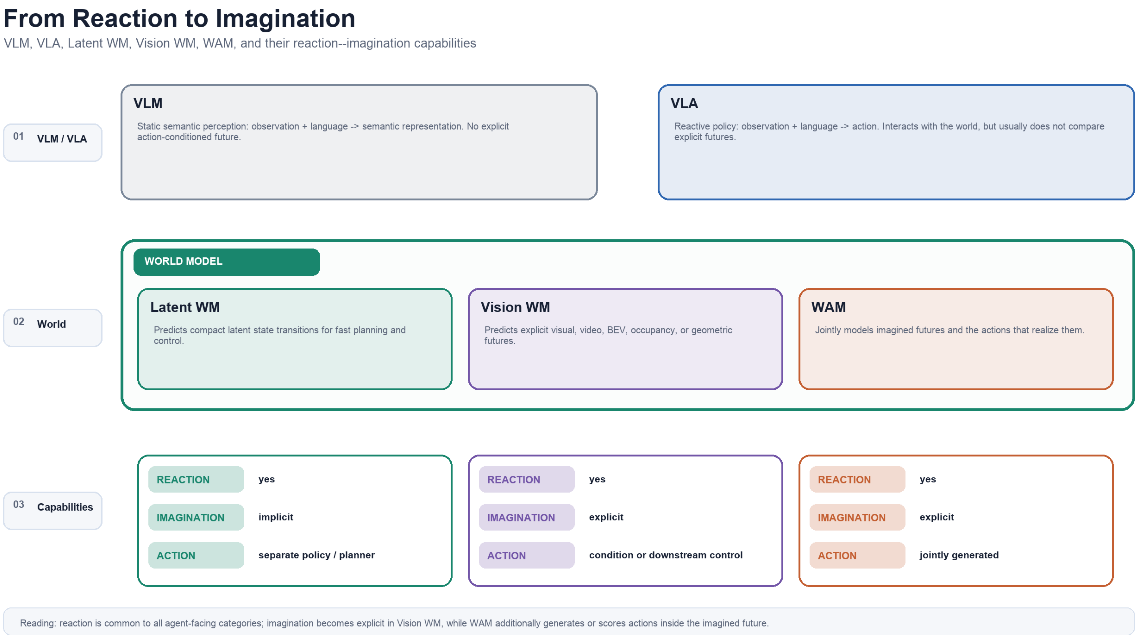

Figure 1: Three-row taxonomy from reaction to imagination. The visible categories are VLM, VLA, Latent WM, Vision WM, and WAM. The capability row distinguishes implicit latent imagination, explicit visual imagination, and explicit imagination coupled to action.

3.1 Latent WM

Latent WM predicts dynamics in a compressed state space. A typical form is

$$z_{t+1} = f_\theta(z_t, a_t), \tag{9}$$

where z_t is a learned belief state, recurrent latent, object state, or search-oriented hidden state. World Models, PlaNet, Dreamer, MuZero, TD-MPC, and TD-MPC2 are representative examples [105, 108, 107, 111, 262, 112].

Its strength is efficiency. A planner can perform many rollouts without rendering pixels. This makes Latent WM attractive for reinforcement learning and high-frequency control. Its weakness is representational risk. If the visual encoder compresses away contact state, object permanence, friction, pose, or a small obstacle, the planner cannot recover that information later. Model exploitation arises when an optimizer finds latent trajectories that look rewarding to the learned model but fail in the physical or rendered environment.

3.1.1 Architecture Evolution: Recurrent Imagination, Search, and MPC

The early *World Models* decomposition used a visual autoencoder, a recurrent dynamics model, and a compact controller: observations were compressed, the recurrent model predicted latent evolution, and the controller learned inside the imagined dynamics [105]. PlaNet replaced this loose composition with a recurrent state-space model and planned online in latent space [108]. Dreamer then amortized planning into actor–critic learning from imagined trajectories; DreamerV2 introduced discrete latent states for Atari-scale domains, while DreamerV3 emphasized a common configuration across diverse tasks [107, 110, 111].

A second lineage learns only the dynamics needed for decision making. MuZero predicts reward, policy, and value in a search-oriented hidden state and combines this model with MCTS, without requiring pixel reconstruction [262]. TD-MPC and TD-MPC2 combine a task-oriented latent transition with temporal-difference value learning and model-predictive control, targeting scalable continuous control [113, 112]. These systems illustrate a key design axis: reconstructing observations improves auditability, while value-equivalent or task-oriented latents can spend capacity only on control-relevant variables.

Model	Developers / Institution	Core Mechanism	Scenario	Main Advantage	Core Limitation
Dreamer V1–V3 [107, 110, 111]	Google / DeepMind research lineage	RSSM belief dynamics; actor–critic learning from imagined latent trajectories	High-frequency RL and continuous/discrete control	High sample efficiency without pixel decoding during policy learning	Planner or policy can exploit model error; compression may erase geometry and contact
MuZero [262]	DeepMind	Learned representation, dynamics, reward, policy, and value combined with MCTS	Board games, Atari, and search-oriented control	Predicts only decision-relevant quantities rather than reconstructing observations	Hidden state is difficult to interpret and need not preserve global spatial detail
TD-MPC2 [112]	Hansen, Su, and Wang; UC San Diego research team	Decoder-free implicit world model, temporal-difference value learning, and local latent trajectory optimization	Multi-task continuous control across domains, embodiments, and action spaces	Strong scaling with a shared configuration and fast short-horizon MPC	Task-oriented latents can underrepresent fine visual geometry and remain exploitable off support
I-JEPA / V-JEPA [12, 23]	Meta AI	Non-generative joint-embedding prediction in semantic feature space	Self-supervised image/video representation and predictive pretraining	Avoids spending capacity on unpredictable pixel detail; robust abstract features	Does not directly render futures or provide an action-conditioned planning interface by itself

Table 3: Representative latent and predictive-representation systems. JEPA is included as a foundation for latent prediction, not automatically as a strict agent-facing world model: its status depends on whether actions condition an exposed future-state interface.

The comparison exposes two latent-design philosophies. Dreamer and TD-MPC2 preserve enough transition structure for repeated control, whereas MuZero preserves only quantities needed by search. JEPA goes further toward representation-first prediction and deliberately avoids pixel reconstruction. This progression improves efficiency, but it also makes evaluation increasingly dependent on downstream control, uncertainty, and adversarial planning tests because visual reconstruction can no longer reveal what the state discarded.

3.1.2 Model Exploitation in Learned Latent Space

Let $\hat{J}_\theta(z_t, a_{t:t+H-1})$ be predicted return and $q_{\mathcal{D}}(z)$ the latent density induced by training data. An unconstrained planner solves

$$\hat{a}_{t:t+H-1} = \arg\max_a \hat{J}_\theta(z_t, a), \tag{10}$$

so optimization pressure actively searches for regions where prediction error is favorable. If high predicted reward coincides with low density, the chosen trajectory may be a model loophole rather than a feasible plan. A risk-aware alternative is

$$\arg\max_a \; \mathbb{E}[\hat{J}_\theta] - \lambda_u \sum_{k=1}^{H} U_\theta(z_{t+k}) - \lambda_d \sum_{k=1}^{H} \big[-\log q_{\mathcal{D}}(z_{t+k}) \big], \tag{11}$$

where U_θ can be ensemble disagreement or posterior uncertainty. Short MPC horizons, periodic re-observation, conservative value estimates, density penalties, and executable checks all reduce exploitation, but they trade planning freedom for reliability.

3.2 Vision WM

Vision WM predicts visible or geometric futures directly. Its output space can be RGB video, video tokens, BEV occupancy, point clouds, depth, scene flow, or a hybrid visual-geometric representation:

$$\hat{I}_{t+1:t+k} = g_\theta(I_t, A_{t:t+k-1}, C). \tag{12}$$

Action-conditioned video prediction, Sora-like video generation, Genie, GameNGen, GAIA-1, DriveDreamer, and related driving world models all belong to this route when their futures are controllable or action-conditioned [234, 34, 35, 299, 127, 315].

Its strength is auditability and semantic richness. A generated visual future can be inspected by humans, used for data generation, or inserted into simulators and evaluation pipelines. Its weakness is the gap between fidelity and physics. A video can look smooth and realistic while violating contact, support, reachability, object permanence, topology, or metric geometry. For agents, perceptual realism is not enough; the future must be causally and physically usable.

3.2.1 Generative Video and Flow/Diffusion Routes

Token-autoregressive models factor future visual tokens sequentially, whereas diffusion or flow-matching systems transform noise into a future video trajectory conditioned on history and controls. A simplified diffusion objective is

$$\mathcal{L}_{\text{video}} = \mathbb{E}_{t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, h, a, c, t)\|_2^2\right], \tag{13}$$

where x_t is a noised video latent. Sora foregrounds scalable visual simulation, Genie introduces controllable interactive environments through latent actions, and GAIA-1 and DriveDreamer condition driving futures on structured context [34, 35, 127, 315]. Flow matching replaces iterative noise prediction with a learned vector field, but the deployment question remains the same: how many visual tokens, integration or denoising steps, and context frames are required before the future is useful for control?

3.2.2 Structured 3D, BEV, and Occupancy Routes

Driving systems expose why pixels alone are insufficient. Multi-camera observations must be lifted into a common metric frame where lanes, occupancy, velocity, occlusion, and collision are measurable. BEVFormer constructs spatiotemporal BEV representations; OccFormer and SurroundOcc predict 3D semantic occupancy; UniAD and VAD connect structured scene representations to planning [185, 377, 322, 130, 141]. Generative driving models such as GAIA-1 and DriveDreamer add visually rich futures, but a safety-facing stack still needs an occupancy or trajectory projection that can be checked.

A structured visual world model can therefore predict

$$p_\theta(O^{\text{occ}}_{t+1:t+H}, X^{\text{agents}}_{t+1:t+H}, I_{t+1:t+H} \mid I_{\le t}, M, A_{t:t+H-1}), \tag{14}$$

where M is map context, O^{occ} is occupancy, and X^{agents} denotes dynamic-agent state. The visual output supports inspection; occupancy and trajectories support metric verification.

3.2.3 Perceptual Realism Is Not Physical Usability

Perceptual metrics reward distributional similarity in appearance and motion. They do not prove that an action caused the predicted change, that a hand applied a feasible wrench, that topology was preserved, or that 2D image motion corresponds to a collision-free 3D trajectory. The gap appears as contact penetration, unsupported objects, disappearing identities, inconsistent multi-view geometry, wrong velocity under occlusion, or futures that cannot be reached by the available actuator. A usable Vision WM therefore needs a layered contract: perceptual plausibility, temporal identity, geometric consistency, action consequence, and executable feasibility.

Model	Developers / Institution	Core Mechanism	Scenario	Main Advantage	Core Limitation
GAIA-1 [127]	Wayve	Autoregressive generative driving model conditioned on video, text, and action context	Camera-level autonomous-driving simulation and policy analysis	Rich semantic and spatial driving futures with controllable context	Visual continuity does not certify collision, contact, or closed-loop safety
Genie [35]	Google DeepMind	Latent-action model plus spatiotemporal video-token dynamics learned from unlabeled video	Interactive and playable 2D environments; later extensions target richer worlds	Discovers control without action labels and turns passive video into an interactive environment	Generation cost and long-horizon identity, topology, and rule drift
Drive-Dreamer [315]	Wang et al. research collaboration	Diffusion world model with two-stage learning of structured traffic constraints and future states	Controllable real-world driving-video generation and policy support	Explicit boxes/maps/traffic structure improve controllability and topology	Depends on structured annotations and still requires metric closed-loop verification
Sora [34]	OpenAI	Large spatiotemporal diffusion-transformer video generator	Open-domain video generation and broad visual simulation priors	Strong appearance, motion, and scene-composition priors at scale	Primarily open-loop without a native executable action interface; intuitive physics is not guaranteed mechanics

Table 4: Representative Vision WM systems and adjacent video simulators. Classification follows the primary output interface: a model can supply valuable visual dynamics while remaining outside the strict agent-facing set when it lacks an exposed action-conditioned control loop.

The table separates *visual prior* from *agent-facing simulator*. Sora contributes broad spatiotemporal priors but is not, by itself, a closed-loop control model under our operational test. Genie moves further by discovering latent controls; GAIA-1 and DriveDreamer add driving context and structure. None eliminates the need to project visual futures into occupancy, trajectories, contact, or another executable state before safety claims are justified.

3.3 WAM

WAM integrates future prediction and action generation in one joint space. Instead of conditioning on a fixed action sequence or outputting only the next action, a WAM models co-trajectories:

$$P_\theta(V_{t+1:t+k}, A_{t:t+k} \mid H_t, G, C), \tag{15}$$

where V denotes future visual, latent, BEV, tactile, or geometric tokens; A denotes action trajectories; H_t denotes history; and G, C denote goal and context. This is the conceptual move behind recent visual world-action systems such as Cosmos-style physical AI world models, WorldDiT, T-TWAM, GeoSem-WAM, GeoWorldAD, and related WAM work [1, 312, 291, 218, 373].

The key idea is causal symmetry. Forward dynamics asks what future follows from an action. Inverse dynamics asks which action realizes a desired future. A WAM attempts to make these two constraints regularize each other. This is more agent-facing than passive video generation and more predictive than reactive VLA. **Constraint leakage** occurs when forward and inverse components assign high joint likelihood or low cycle loss to a visual–action trajectory that violates a hard constraint absent from their training objective, such as collision, kinematic reachability, contact mechanics, force, or a safety rule. Unlike perceptual realism failure, which concerns an invalid predicted world, constraint leakage specifically survives agreement between predicted world and generated action.

3.3.1 Forward–Inverse Causal Symmetry

Let F_θ predict future features from actions and I_ψ infer actions from a current and desired future state:

$$\hat{v}_{t+1:t+H} = F_\theta(v_{\le t}, a_{t:t+H-1}, g), \tag{16}$$

$$\hat{a}_{t:t+H-1} = I_\psi(v_{\le t}, v^*_{t+1:t+H}, g). \tag{17}$$

A joint token model can train both directions with

$$\mathcal{L}_{\text{WAM}} = \lambda_f \mathcal{L}_{\text{forward}} + \lambda_i \mathcal{L}_{\text{inverse}} + \lambda_c \|I_\psi(v_t, F_\theta(v_t, a)) - a\| + \lambda_p \mathcal{L}_{\text{physical}}, \tag{18}$$

where cycle consistency discourages a future that cannot explain its generating action, and $\mathcal{L}_{\text{physical}}$ penalizes geometry, contact, or kinematic violations when labels or executable checks are available. Mutual attention or joint diffusion allows future tokens to condition action tokens and vice versa. The benefit is regularization across causal directions; the danger is collusion, where both heads agree on the same physically wrong latent convention.

3.3.2 Representative 2026 Design Differences

The recent systems in Table 5 should be read as different attempts to close the representation–action gap, not as directly comparable benchmark entries. Cosmos 3 broadens the conditioning and output interface toward omnimodal physical AI; WorldDiT explicitly unifies world and action modeling in a diffusion architecture; T-TWAM makes tactile signals native to contact-rich world-action modeling; GeoSem-WAM introduces geometry and semantics as named organizing constraints [1, 312, 291, 218].

Model	Developers / Institution	Core Mechanism	Scenario	Main Advantage	Core Limitation
Cosmos 3 [1]	NVIDIA and collaborators	Omnimodal physical-AI world-model interface spanning visual futures and action-relevant context	Robotics, autonomous driving, and physical-AI data/simulation stacks	Broad multimodal priors and a shared foundation for diverse embodiments	Large token, training, and inference budgets; hard physical validity still needs executable checks
World-DiT [312]	Wang, Praveen, Roy, and Villagra	One diffusion transformer generates continuous action chunks and predicts normalized RGB patches from future camera frames	Robot manipulation on LIBERO suites	Couples action generation and visual prediction without a large pretrained VLM action backbone	Joint likelihood can hide constraint leakage; future patches do not certify kinematic or contact feasibility
T-TWAM tactile-native line [291]	NeoteAI Team / Fudan TEAI Team	Visuo-tactile joint training, NeoForce contact representation, contact-event staging, and asymmetric Mixture-of-Transformers experts	Contact-rich, multi-stage manipulation across embodiments	Predicts vision, touch, and action while retaining contact events hidden from vision alone	Long-horizon cross-modal errors can compound; real-time gains rely on asymmetric experts and substantial tactile data
GeoSem-WAM [218]	Ma et al. research team	Geometry- and semantic-aware world-action representation	Spatially grounded embodied control	Makes geometric feasibility and semantic task meaning explicit in joint modeling	Structured priors improve grounding but add representation and supervision complexity

Table 5: Representative WAM systems. The systems are heterogeneous and are compared by interface and mechanism rather than as a single leaderboard; institution labels are used only when directly established, otherwise the author team is reported.

Across these WAMs, "generation is decision" has three distinct meanings: shared world/action tokens, one joint denoising backbone, or synchronized modality-specific experts. The stronger coupling can reduce mismatch between a separately trained predictor and controller, but it also creates a coupled error channel: visual, tactile, and action predictions may reinforce one another while violating a constraint absent from the loss. This is why the hybrid verifier in Section 6 is complementary to, rather than replaced by, end-to-end WAM training.

Together, the three route-specific matrices replace a single coarse comparison: they preserve developer provenance, mechanism, scenario, advantage, and limitation while allowing multi-interface systems to be discussed without forcing one mutually exclusive row. The next section shifts from model-level comparison to corpus-level technology evolution and resource trade-offs.

4 Technology Evolution: From Latent Rollout to Joint World–Action Modeling

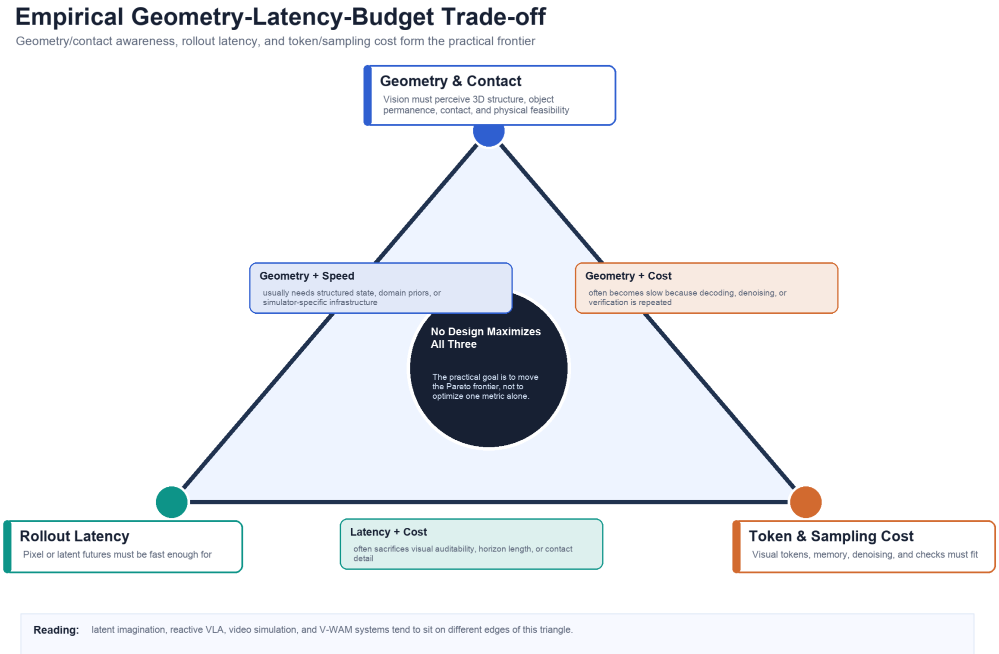

Figure 2: Empirical geometry–latency–budget trade-off. Each category occupies a different region of the observed frontier; no impossibility theorem is claimed. Latent WM favors latency and budget, Vision WM favors inspectable detail, and WAM adds action coupling while still requiring hard verification.

The taxonomy is not only conceptual; it also describes a historical progression. Early work prioritized compact dynamics and cheap imagined rollout. Subsequent representation-learning and video-generation systems increased visual fidelity and temporal range. Robotics, driving, and interactive environments then introduced explicit controls, structured geometry, and task-facing evaluation. The recent frontier couples future prediction to action generation and executable verification.

The sequence should not be read as one architecture replacing another. Latent dynamics, predictive representations, video generation, VLA policies, structured driving state, and WAMs continue to coexist. The trend is an accumulation of capabilities: compression enables rollout, visual generation exposes futures, action conditioning makes futures controllable, and joint modeling links imagination to decisions. Figure 2 places this progression against the geometry, latency, and compute frontier before the bottleneck analysis.

5 Technical Bottlenecks: The Geometry–Latency–Budget Trade-off Triangle

The empirical trade-off triangle remains central, but it is not an impossibility theorem. We conjecture, without proving a lower bound, that current neural architectures exhibit a Pareto surface among **geometry/contact awareness**, **low rollout latency**, and **affordable compute**: no system in the curated corpus demonstrates all three at deployment-grade levels under a common benchmark. Figure 2, placed at the opening of Section 4, visualizes this empirical reading. The constrained optimization below selects a point on the frontier; it does not remove the trade-off.

Latent WM chooses the latency-budget edge. It can roll out futures cheaply in compact state, but the compression process can remove the very geometric or contact details that make a plan executable. Vision WM chooses the fidelity edge. It can render rich futures, but denoising, high-resolution tokens, long contexts, and geometry lifting increase latency and compute. WAM tries to move the frontier by generating future state and action together, but it inherits both sides of the problem: joint generation increases modeling burden, and a statistically coherent visual-action sequence can still be physically invalid.

Let G denote geometry/contact awareness, τ denote rollout latency, and B denote token, sampling, and verification budget. A useful world model should maximize G while keeping τ and B below deployment thresholds:

$$\max_{\theta} G_\theta \quad \text{subject to} \quad \tau_\theta \leq \tau^*, \; B_\theta \leq B^*. \tag{19}$$

Every known improvement path stresses this objective. Higher resolution, longer horizon, richer memory, stronger multimodal conditioning, 3D geometry, diffusion sampling, ensembles, or external verifiers can raise G, but they also tend to increase τ or B. Aggressive compression and direct policies reduce τ and B, but they often hide physical state.

5.1 Constraint Solving and Frontier-Shifting Strategies

The constrained objective can be trained with a Lagrangian relaxation,

$$\mathcal{L}(\theta, \lambda_\tau, \lambda_B) = -G_\theta + \lambda_\tau(\tau_\theta - \tau^*) + \lambda_B(B_\theta - B^*), \tag{20}$$

with nonnegative dual variables updated online:

$$\lambda_\tau \leftarrow [\lambda_\tau + \eta_\lambda(\tau_\theta - \tau^*)]_+ \,, \tag{21}$$

$$\lambda_B \leftarrow [\lambda_B + \eta_\lambda(B_\theta - B^*)]_+ \,. \tag{22}$$

This turns a fixed weighted sum into a deployment-aware controller: measured latency or budget violations increase their own penalty. Because G, τ, and B may be nondifferentiable at execution time, practical systems combine surrogate training losses with measured wall-clock latency, token count, denoising steps, peak memory, and verifier calls.

Four frontier strategies recur. **Adaptive horizon** uses short latent rollout for routine states and lengthens imagination near irreversible decisions. **Coarse-to-fine rollout** evaluates many actions in a cheap latent or low-resolution model, then renders and verifies only the top candidates. **Early-exit generation** stops denoising or decoding once action ranking is stable, rather than requiring publication-quality pixels. **Risk-triggered verification** allocates collision, IK, contact, or map checks to candidates with high uncertainty, low latent density, or small safety margin. These strategies do not remove the triangle; they spend expensive fidelity only where it can change the action.

At inference time, a resource-aware planner can solve

$$a^* = \arg\max_{a \in \mathcal{A}} \left[\hat{J}(a) - \beta_u U(a) - \beta_v R_{\text{verify}}(a) \right] \quad \text{s.t.} \quad \sum_{a \in \mathcal{A}} (c_{\text{roll}}(a) + c_{\text{verify}}(a)) \le B^*, \tag{23}$$

where verification risk is infinite for a hard rejection and graded for repairable violations. The resulting policy jointly chooses an action and decides how much computation is justified before committing it.

Category	Triangle Choice	Typical Failure	Needed Correction
Latent WM	Low latency + low budget	Model exploitation, hidden contact error, latent blind spots	Density penalties, uncertainty-aware MPC, object memory, executable checks
Vision WM	Geometry/visual detail + auditability	Fidelity is not physics; realistic video violates constraints	Geometry-aware generation, contact metrics, downstream task evaluation
WAM	Joint prediction-action coupling	Constraint leakage across visual and action tokens	Typed verification, repair loops, physical and kinematic shields
VLA	Low latency + direct action	Reactive blindness under occlusion or irreversible actions	Future rollout heads or WAM augmentation

Table 6: The empirical trade-off triangle explains why each taxonomy category exposes a different failure mode; it is a comparative lens, not a proven lower bound.

6 Future Path: Hybrid Physics-Neural World Models

The most practical path is not a pure neural model that learns all physics from pixels. It is a hybrid architecture in which a neural world model proposes futures and an executable verifier checks constraints. The neural side supplies semantic breadth, generative diversity, and open-world visual priors. The executable side supplies collision checks, kinematics, map constraints, force limits, contact validity, and rule enforcement.

A minimal hybrid loop is

$$(\hat{s}_{t+1:t+H}, \hat{a}_{t:t+H}, u_t) = F_\theta(h_t, g, \ell, c), \tag{24}$$

$$(\rho_t, s^{\text{safe}}_{t+1:t+H}) = C(\hat{s}_{t+1:t+H}, \hat{a}_{t:t+H}, E), \tag{25}$$

where F_θ is the neural proposer, u_t is uncertainty, C is a physics/geometric verifier or executable shield, E is an environment substrate, and ρ_t is a typed approval, rejection, or repair code. Typed rejection is crucial. A scalar penalty only says that a proposal is bad; a typed rejection says whether the failure is collision, unreachable pose, rule violation, geometry mismatch, contact inconsistency, missing state, or verifier timeout.

6.1 Operational Propose–Project–Verify–Repair–Commit Loop

The workflow is executed over a candidate set rather than one deterministic future:

1. **Propose.** Sample K visual-action trajectories $\xi_i = (\hat{s}^i_{t+1:t+H}, \hat{a}^i_{t:t+H-1})$ together with epistemic uncertainty, predicted utility, latent density, and required state fields.
2. **Project.** Convert each neural trajectory into executable state: robot joint paths and contact frames, driving BEV/occupancy and agent tracks, or simulator entities and event state. Projection must report missing variables rather than silently impute them.

3. **Verify.** Apply broad-phase collision tests, inverse kinematics, joint/force limits, map and traffic rules, contact consistency, occupancy safety margins, and task invariants. Cheap checks run first; expensive simulation is reserved for survivors.
4. **Repair.** Use the typed rejection code to constrain local trajectory optimization, replace an invalid waypoint, tighten the generation condition, or resample. Repair is bounded by retry and latency budgets.
5. **Commit.** Select the highest-utility approved trajectory, execute only its next receding-horizon segment, observe the actual postcondition, and return drift plus rejection traces as hard negatives.

For candidate i, a deployable selector can use

$$S_i = \hat{J}_i - \beta_u u_i - \beta_d D_i - \beta_c C_i, \tag{26}$$

where D_i is predicted-to-executed state discrepancy and C_i is verification cost; any hard rejection sets $S_i = -\infty$. Receding-horizon commitment limits damage from long-rollout error because the model is reconditioned on real observations after each short segment.

Figure 3 makes the execution contract explicit. The important loop is not a one-way neural pipeline: verification failures are typed, routed to repair, and retained as evidence for future proposal and calibration updates.

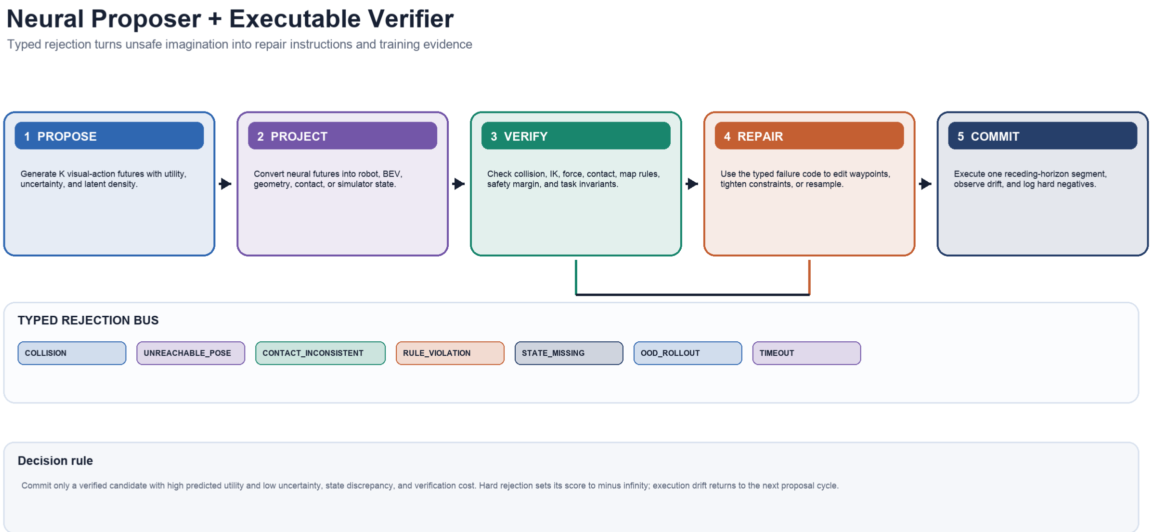

Figure 3: Hybrid proposer–verifier workflow. Safety improves only when executable checks return typed failure information to repair and learning; a scalar plausibility score cannot distinguish collision, reachability, contact, missing-state, distribution-shift, and latency failures.

6.2 Typed Rejection Codes as a Learning Interface

Typed codes make safety decisions auditable, route candidates to the correct repair operator, and create supervised hard negatives for subsequent training. A compact schema is shown in Table 8.

To close the learning loop, store each tuple $(h, a, \hat{s}', \rho, s'_{\text{exec}})$ in a rejection buffer and optimize a code-conditioned margin loss that ranks a repaired or verified trajectory above the rejected proposal. For example, a WAM may satisfy forward–inverse cycle consistency for a visually smooth grasp, yet receive `UNREACHABLE_POSE` from IK; the update must penalize the missing kinematic constraint rather than merely lowering generic video likelihood. Code-stratified validation then reports recurrence and repair success for each failure type.

Step	Owner	Contract	Failure Handling
`propose`	Neural WM or WAM	Return K candidate visual-action futures with uncertainty and required state variables	Abort if uncertainty is high or state variables are missing
`project`	Interface layer	Convert visual/latent future into executable geometry, BEV, robot, or simulator state	Request repair if projection loses required constraints
`verify`	Physics or executable shield	Check collision, reachability, contact, map rules, task invariants, and safety envelope	Return typed rejection code
`repair`	Joint loop	Resample or locally edit the proposal under typed constraints	Stop after repeated failure or latency overrun
`commit`	Agent or simulator	Execute only verified action plans and log postcondition drift	Roll back or hand off if observed state diverges

Table 7: Minimal API contract for a hybrid physics-neural world model.

This hybrid path preserves the strengths of each route. Latent WM provides efficient internal rollout; Vision WM provides inspectable counterfactual futures; WAM couples desired futures with action trajectories; VLM/VLA components provide semantics and action priors. The verifier prevents these neural components from silently turning missing physics into plausible pixels or fluent action tokens.

Recent systems make pieces of this loop concrete. CheckVLA uses an action-conditioned world model for execution-time verification in long-horizon mobile manipulation, while hybrid world-model/VLA work in driving illustrates how learned and structured components can share the decision loop [204, 358]. The remaining engineering problem is verifier coverage: a checker can reject only violations represented in its executable state.

Code	Detection	Immediate Response	Training Feedback
COLLISION	Geometry overlap or unsafe occupancy envelope	Reroute waypoint or reject trajectory	Contrast safe and unsafe spatial futures
UNREACHABLE_POSE	IK failure, joint limit, or disconnected configuration	Project goal into reachable set	Penalize inverse-dynamics actions without feasible realization
CONTACT_INCONSISTENT	Missing support, penetration, slip, or incompatible tactile event	Resample contact phase or force profile	Align visual, tactile, and action tokens at contact transitions
RULE_VIOLATION	Map, traffic, task, or game invariant failure	Add symbolic constraint and replan	Train rule-conditioned rejection or value head
STATE_MISSING	Projection lacks mass, velocity, identity, map, or object state	Acquire observation or abstain	Mark hidden variable as required model output
OOD_ROLLOUT	Ensemble disagreement or low latent density	Shorten horizon or use conservative controller	Add execution trace to uncertainty calibration set
TIMEOUT	Generation or verification exceeds τ^*	Fall back to cached or reactive safe policy	Train compute-aware early exit and candidate pruning

Table 8: Typed rejection codes for closing the loop between executable verification, repair, and world-model training.

7 Benchmarks, Metrics, and Evaluation

Evaluation is asymmetric across the corpus. Broad-support work in video generation and representation learning often reports reconstruction loss, PSNR, SSIM, LPIPS, FVD, or feature-prediction accuracy. Strict agent-facing world models must additionally report whether prediction improves action, remains stable under feedback, preserves physical constraints, and recognizes when it should abstain. A single perceptual metric cannot bridge this gap.

Figure 4 organizes this asymmetry as a ladder. Every higher level can expose a model that passes the lower levels: perceptually convincing video may fail geometry, action causality, closed-loop control, rare-event calibration, or deployment latency.

7.1 Open-Loop versus Closed-Loop Evaluation

Open-loop evaluation conditions on a fixed recorded history and compares predicted futures with held-out observations. It is reproducible and diagnostic for appearance, motion, and short-horizon state error. However, the model never changes the data distribution: its predicted state is not fed to a planner whose action changes the next observation. Closed-loop evaluation repeatedly performs

$$\hat{s}_{t+1} = F_\theta(h_t, a_t), \qquad a_t = \pi(\hat{s}_{t+1:t+H}), \qquad o_{t+1} \sim E(\cdot \mid o_t, a_t), \tag{27}$$

so model error changes action and action changes future input. The useful quantities are therefore cumulative state drift, policy regret, task success, safety violations, recovery, and calibration under distribution shift.

A horizon-normalized rollout error can be written as

$$E_H = \frac{1}{H}\sum_{k=1}^{H} d(\hat{s}_{t+k}, s_{t+k}), \tag{28}$$

but closed-loop utility requires comparing policies:

$$\mathrm{Regret}_H = J(\pi^*, E) - J(\pi_{F_\theta}, E), \tag{29}$$

Figure 4: Evaluation ladder for agent-facing world models. Open-loop perceptual fidelity is necessary but insufficient: a strict world-model claim also requires geometric state consistency, causal action coupling, closed-loop utility, calibrated safety, and deployment-feasible compute.

where π_{F_θ} selects actions using the learned model. Because π^* is normally unavailable, empirical regret should use a declared reference: an environment oracle when available, otherwise the strongest fixed comparator evaluated on the same task seeds and budget. A model can have low open-loop E_H yet high regret if its small errors occur near contacts, collisions, or irreversible decisions.

Closed-loop evaluation protocol. Input a frozen world model F_θ, planner P, executable environment E, task/seed set $\mathcal{T}$, horizon H, and latency/compute budget B^*. For each task, reset the same seed for all methods; repeatedly generate and score candidate actions, execute only the selected receding-horizon action, and log predicted/observed state discrepancy, action, typed verifier result, fallback or human intervention, task progress, collision/rule events, and wall-clock/token cost. Report paired confidence intervals for success, safety-event rate, reference-policy regret, calibration, interventions per episode, and budget violations. An intervention is counted whenever a human, safety fallback, or executable shield overrides the model-selected action; results must separate these three sources. For risk calibration, define a failure event before evaluation and compute NLL/ECE or coverage–risk over the model's predicted probability of that event, not over generic pixel likelihood.

7.2 Intervention, Rare-Event, and Safety Evaluation

Counterfactual evaluation should hold history fixed and vary the intervention. What-If World explicitly targets causal world-model behavior, while KineBench targets kinematic grounding without relying on inverse-dynamics labels [39, 205]. A paired-action test queries $\hat{s}'_1 = F(h_t, a_1)$ and $\hat{s}'_2 = F(h_t, a_2)$, then measures whether $d(\hat{s}'_1, \hat{s}'_2)$ and each prediction's error against executed outcomes reflect the known action difference. Action swaps, delays, impossible actions, hidden-state changes, and exogenous perturbations should change only causal descendants while preserving unaffected scene variables. This test operationalizes the taxonomy boundary without claiming that every latent temporal feature is a world model.

Safety-critical events are rare under natural logs, so average error is a weak estimator. Driving evaluation should oversample near-collision, occlusion, cut-in, wrong-way, and adverse-weather scenarios; manipulation evaluation should oversample unstable grasps, thin obstacles, slip, force-

Evaluation Layer	Representative Metrics	What It Tests	What It Misses Alone
Perceptual open loop	PSNR, SSIM, LPIPS, FVD	Appearance and distributional video similarity	Causality, reachability, contact, policy utility
State/geometric open loop	ADE/FDE, occupancy IoU, depth, flow, object persistence	Metric and temporal state prediction	Planner-induced distribution shift
Action coupling	Inverse-dynamics accuracy, action-conditioned consistency, cycle error	Whether actions explain predicted change	Executability under hard constraints
Closed-loop control	Return, task success, planning score, collision rate, intervention count	Whether imagination improves acting under feedback	Rare failures without targeted stress tests
Risk and calibration	NLL, ECE, coverage–risk, abstention utility, ensemble disagreement	Whether confidence tracks error and OOD state	Safety of unmodeled constraints
System efficiency	End-to-end latency, tokens, denoising steps, memory, verifier calls	Whether the model fits the deployment budget	Physical quality unless paired with task metrics

Table 9: A layered evaluation protocol linking perceptual quality to closed-loop action utility, safety, and compute.

limit violations, and irreversible drops. Under proposal distribution $q(x)$, rare-event risk can be estimated by importance weighting,

$$\hat{p}_{\text{fail}} = \frac{1}{N} \sum_{i=1}^{N} \frac{p(x_i)}{q(x_i)} \mathbf{1}[\text{failure}(x_i)], \qquad x_i \sim q, \tag{30}$$

provided support and weights are controlled. Stress suites should report both weighted risk and raw failure taxonomy, because a single scalar hides whether failures come from perception, rollout, action coupling, or verifier coverage. RoboTrustBench, SafeVLA-Bench, and physical-grounding benchmarks exemplify the move toward trust, success–safety gaps, and physics-aware evaluation [172, 80, 190].

7.3 Benchmark and Open-Framework Map

Table 10 connects the corpus treemap's major domains to evaluation assets. "Open framework" means an executable environment, dataset, or benchmark interface useful for reproducibility; it does not imply that every cited world model releases complete training code.

Domain	Benchmarks / Frameworks	Required World-Model Outputs	Strict Evaluation Target
General / latent control	Meta-World, model-based control suites [359]	Latent state, reward/value, uncertainty	Return, sample efficiency, model exploitation, OOD planning
Robotics manipulation	RLBench, CALVIN, LIBERO, RoboSuite [136, 223, 195, 221]	Object/contact state, future images, actions, task progress	Long-horizon success, contact feasibility, recovery, cross-task transfer
Robot data and generation	Open X-Embodiment, BridgeData V2, RoboGen [63, 305, 316]	Cross-embodiment tokens and generated trajectories	Transfer, synthetic-data utility, embodiment shift
Autonomous driving	CARLA, nuPlan, VISTA 2.0 [72, 38, 10]	BEV/occupancy, agent trajectories, ego plan, uncertainty	Closed-loop progress, collision, rules, rare-event robustness
Video world models	MBench, What-If World, WorldArena 2.0 [372, 39, 266]	Persistent visual state, interventions, controllable futures	Memory, causal response, multi-platform interaction
Physical / kinematic grounding	KineBench, PhyGround, RoboTrustBench [205, 190, 172]	Pose, contact, geometry, action-conditioned state	Reachability, physics, calibration, trustworthy abstention

Table 10: Representative benchmark and open-framework mapping for the major corpus domains.

8 Applications and Future Challenges

The taxonomy becomes useful when it changes system design. The following cases show how prediction space, action coupling, and verification requirements differ across two high-risk applications.

8.1 Case Study I: BEV/Occupancy Prediction and Planning in Autonomous Driving

A driving stack begins with synchronized multi-camera observations and ego state. BEVFormer-style lifting produces a shared bird's-eye representation; occupancy models represent free, occupied, and uncertain 3D cells; a world model predicts dynamic agents and occupancy under candidate ego actions; a planner scores progress, comfort, collision, and rule compliance [185, 377, 322]. UniAD and VAD exemplify planning-oriented integration, while GAIA-1 and DriveDreamer contribute generative visual futures [130, 141, 127, 315].

The practical architecture is hybrid:

$$I_{\le t}^{1:N} \to Z_t^{\mathrm{BEV}} \to \{\hat{O}^i_{t+1:t+H}, \hat{X}^i_{t+1:t+H}\}_{a_i} \to \text{rule/collision verify} \to a^*. \tag{31}$$

The video or camera-view future supports human audit and sensor realism; BEV/occupancy supports metric collision checks; the trajectory head supports control. Closed-loop evaluation in CARLA, nuPlan, or VISTA should vary occlusion, agent intent, map topology, and weather rather than replaying only logged actions [72, 38, 10]. The unresolved issue is long-tail calibration: the system must know when an occluded or behaviorally unusual agent invalidates its predicted occupancy.

8.2 Case Study II: Tactile–Visual WAMs for Contact-Rich and Bimanual Manipulation

Hand and dual-arm manipulation reveal variables that vision cannot reliably recover: contact onset, normal force, slip, compliance, internal grasp force, and occluded finger–object geometry.

A tactile–visual WAM should maintain modality-specific encoders but synchronize them around contact events:

$$p(v_{t+1:t+H}, q_{t+1:t+H}, c_{t+1:t+H}, a_{t:t+H-1} \mid v_{\leq t}, q_{\leq t}, c_{\leq t}, g), \tag{32}$$

where v is visual state, q is proprioception, and c is tactile/contact state. T-TWAM and Dream-Tac explicitly foreground tactile-native or unified tactile world-action modeling, while contact-image and cross-embodiment action-conditioning work provides complementary contact representations [291, 209, 328, 14].

For bimanual tasks, the verifier must check each arm's IK and collision constraints plus coupled invariants: force closure, relative object pose, handover timing, and whether one arm occludes or blocks the other. Evaluation should report task completion, contact-state F1, slip prediction, peak force violation, bimanual synchronization error, recovery after perturbation, and real execution success. CALVIN, LIBERO, RLBench, and RoboSuite provide reusable task environments, but tactile/contact channels and dual-arm physics often require domain-specific extensions [223, 195, 136, 221].

8.3 Cross-Domain Future Challenges

Five challenges remain common. First, **hidden-state identification**: video does not uniquely reveal physical parameters or agent intent. Second, **long-horizon memory**: generated futures drift in identity, topology, and rules. Third, **causal action data**: demonstrations cover successful behavior better than counterfactual failures. Fourth, **verifier coverage**: hard checks are precise but narrow. Fifth, **evaluation standardization**: perceptual, physical, control, safety, and compute metrics are rarely reported together. Progress should therefore be measured by movement of the complete system frontier, not by a single visual or task score.

9 Conclusion

This reconstruction defines world models by two criteria: interactivity and predictive imagination. This reframing clarifies why VLMs and many VLAs should be treated as neighboring categories rather than automatically counted as world models. VLMs perceive and describe; reactive VLAs act; true world models imagine action-conditioned futures. The probabilistic SSM view further clarifies that imagination must maintain belief and uncertainty under partial observability, while counterfactual use requires intervention support beyond fitting an observational next-state distribution.

Under this boundary, the technical history becomes a clean progression. Latent WM makes imagination cheap but risks losing physical detail. Vision WM makes futures visible but risks confusing visual fidelity with physics. WAM brings prediction and action into one joint space but still needs hard checks for geometry, contact, and safety. The empirical geometry–latency–budget trade-off explains why none of these routes is sufficient alone.

The evaluation consequence is equally important. FVD, reconstruction error, or one-step state accuracy can diagnose a component, but they cannot establish closed-loop utility. A credible evaluation stack must jointly report perceptual/state fidelity, action coupling, task success, rare-event safety, calibration, and end-to-end compute. The driving and tactile-manipulation cases show why structured geometry and contact channels become indispensable exactly where visual plausibility is easiest to mistake for physical validity.

The survey's forward claim is therefore hybrid. The next useful world model is not just a larger video generator or a more fluent robot policy. It is a physics-neural system in which neural models propose diverse semantic futures, resource-aware planners allocate imagination and verification budget, and executable verifiers enforce the constraints required for real action.

Typed rejection and repair turn failure from a terminal scalar penalty into training data. That is the path from reaction to causal predictive imagination.

A Corpus Distribution Treemap and Vision-Role Mapping

The corpus is summarized with a hierarchical treemap rather than a 405-row table. The first layer encodes domain, the second layer encodes primary stack, and the third layer encodes strict or broad inclusion role. In the revised vision-centric framing, the primary stack layer is read as a proxy for the vision model's role: encoder/compressor, perceptual backbone, simulator core, joint visual-action space, geometry bridge, or executable shield.

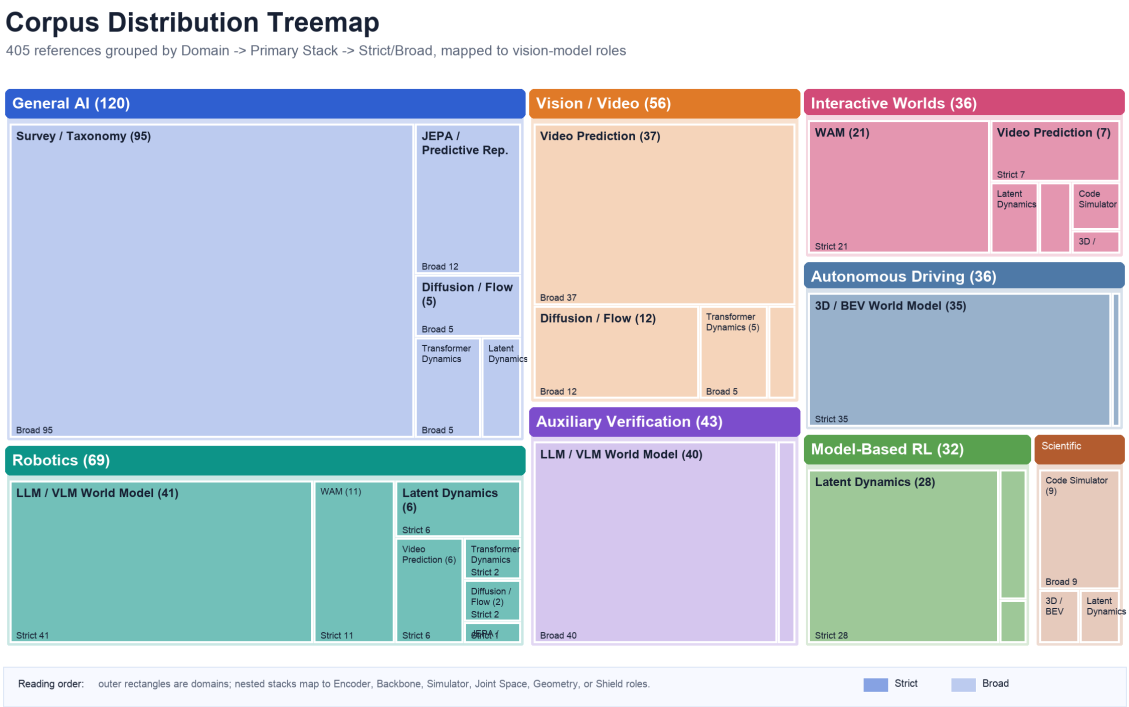

Figure 5: Treemap of the 405-entry corpus. Rectangle area is proportional to paper count; color denotes domain; nested rectangles represent primary stack and strict/broad inclusion role, with primary stacks interpreted through the survey's vision-model role taxonomy.

A.1 Treemap Interpretation

The treemap should be read as a distributional map rather than a bibliography table. The largest domain block is General AI with 120 entries, followed by Robotics with 69 entries. The most frequent stack overall is Survey / Taxonomy, which appears in 95 entries. In the main text, primary stacks are mapped onto vision-model roles rather than treated as final categories: encoder/compressor, perceptual backbone, simulator core, joint visual-action space, geometry bridge, and executable shield. Across the corpus, 173 entries are strict agent-facing world-model works and 232 entries are broad supporting works. This split is important: strict entries define the core evidential base for world-model claims, while broad entries explain the representation, generative modeling, benchmark, and simulation machinery that modern world models reuse.

Domain	Count	Dominant Stack	Role Mix	Reading
General AI	120	Survey / Taxonomy	S:0/B:120	Foundational methods and surveys supply representation, planning, and generative priors.
Robotics	69	LLM / VLM World Model	S:69/B:0	VLA and manipulation works connect multimodal policies with action consequence modeling.
Vision / Video	56	Video Prediction	S:0/B:56	Visual future modeling is mainly broad support unless directly action-conditioned.
Auxiliary Verification	43	LLM / VLM World Model	S:0/B:43	Tool, UI, and code systems provide executable checks around visually grounded plans.
Autonomous Driving	36	3D / BEV World Model	S:36/B:0	BEV and occupancy stacks dominate because safety depends on structured scene state.
Interactive Worlds	36	WAM	S:36/B:0	Game and controllable video systems are small but highly relevant to WAMs.
Model-Based RL	32	Latent Dynamics	S:32/B:0	Latent dynamics remains the strongest strict lineage for planning through imagination.
Scientific Simulation	13	Code Simulator	S:0/B:13	Neural simulators contribute dynamics priors but are usually not agent-facing.

Table 11: Compact reading of the domain layer in the corpus treemap.

A.2 Distributional Observations

First, the corpus is not uniformly distributed across domains. General AI and vision/video occupy large areas because they provide the representation learning, generative modeling, and survey vocabulary used by more specialized world-model systems. Their broad role is therefore expected rather than a weakness: these works define the technical substrate from which agent-facing systems borrow encoders, tokenizers, predictors, diffusion objectives, and evaluation conventions. Second, the strict blocks are concentrated in domains where action consequences are explicit. Model-based reinforcement learning, robotics, autonomous driving, and interactive worlds are visually distinct because their problems naturally expose interventions. In those domains, a prediction is not only a future sample; it is evidence for whether a policy, planner, driving maneuver, manipulation action, or generated action sequence should be executed. Third, technology stacks cluster by the state variables that each domain treats as authoritative. Autonomous driving favors 3D and BEV state because safety depends on geometry, occupancy, route constraints, and multi-agent trajectories. Auxiliary verification favors LLM/VLM and code-simulator stacks because rendered screens, APIs, permissions, files, and execution traces can validate whether a visual plan has unsafe side effects. Model-based reinforcement learning favors latent dynamics because compact imagination remains efficient for return optimization. Fourth, video and diffusion systems are large but mixed. They are crucial for visual simulation and interactive worlds, yet visual realism alone does not make a system an agent-facing world

Primary Stack / Vision Role	Count	Main Domains	Role Mix	Interpretation
Survey / Taxonomy	95	General AI	S:0/B:95	Broad conceptual work that defines where vision models sit in the action loop.
LLM / VLM World Model	81	Auxiliary Verification, Robotics	S:41/B:40	Perceptual-backbone role for reactive VLA and visually grounded screens; non-visual use is treated only as backend verification.
Video Prediction	51	Autonomous Driving, Interactive Worlds, Robotics	S:14/B:37	Simulator-core role when future generation is tied to action, control, or interaction.
Latent Dynamics	42	General AI, Interactive Worlds, Model-Based RL	S:37/B:5	Encoder/compressor role for compact imagination, value learning, and model predictive control.
3D / BEV World Model	38	Autonomous Driving, Interactive Worlds, Scientific Simulation	S:36/B:2	Geometry-bridge role for lifting visual features into BEV, occupancy, and trajectory state.

Table 12: Compact reading of the primary-stack layer in the corpus treemap (part 1).

model. The role layer in the treemap is therefore essential: it separates broad generative foundations from strict systems that condition on actions, evaluate consequences, or close the loop with planning and control. Fifth, the treemap reveals an evaluation asymmetry. The largest broad regions are usually evaluated with reconstruction, likelihood, perceptual quality, or representation metrics, while strict regions need task success, safety, calibration, controllability, and closed-loop robustness. This asymmetry explains why a survey organized only by architecture would miss a central point: the same model category can be evaluated very differently once it becomes part of an acting agent. Sixth, the small size of the WAM and interactive-world blocks should not be interpreted as low importance. These regions mark a frontier rather than a mature literature. Their significance lies in the coupling between future generation and action generation. As more systems move from passive prediction to controllable simulation, these blocks are likely to expand and merge with robotics, digital agents, and video-generation research. Seventh, the broad scientific-simulation block highlights a boundary case. Neural operators, graph simulators, weather models, and physics surrogates learn dynamics, but they are not always agent-facing. They become central to world-model research when their predictions support planning, intervention, control, or verification. This distinction prevents the corpus from treating every predictive physical model as a world model while still acknowledging that scientific simulators provide important design lessons. Finally, the treemap should be interpreted together with the main taxonomy rather than as a replacement for it. Area gives a quick sense of volume, but it does not measure impact, maturity, benchmark difficulty, or deployment risk. A small strict block may be more consequential for safety than a large broad block, and a broad foundation model may enable several future strict systems. The figure is therefore a navigational device: it helps readers see where the literature is concentrated before returning to the technical and evaluative arguments in the main text.

Primary Stack / Vision Role	Count	Main Domains	Role Mix	Interpretation
WAM	32	Interactive Worlds, Robotics	S:32/B:0	Joint-space role because generated futures and generated actions are coupled directly.
Diffusion / Flow	22	General AI, Interactive Worlds, Model-Based RL	S:5/B:17	Simulator-core or V-WAM mechanism for multimodal futures and trajectories.
JEPA / Predictive Rep.	15	General AI, Robotics, Vision / Video	S:1/B:14	Encoder/compressor role for semantic prediction that avoids pixel reconstruction as the only target.
Transformer Dynamics	15	General AI, Model-Based RL, Robotics	S:5/B:10	Tokenization mechanism shared across latent imagination, VLA, simulator-core, and V-WAM routes.
Code Simulator	14	Auxiliary Verification, Interactive Worlds, Scientific Simulation	S:2/B:12	Executable-shield role after visual plans touch rendered state, tools, code, or environment execution.

Table 13: Compact reading of the primary-stack layer in the corpus treemap (part 2).

A.3 Implications for Survey Structure and Evaluation

The distribution also explains why this survey is organized around frameworks and architectural layers rather than around domains alone. A domain-only survey would separate robotics, autonomous driving, digital agents, and games, but the treemap shows that these areas repeatedly reuse the same technical categories: latent dynamics, transformer dynamics, diffusion or flow models, LLM/VLM grounding, code execution, and structured geometry. Conversely, an architecture-only survey would hide the fact that the same stack has different meanings in different domains. Diffusion is a video generator in one block, a trajectory prior in another, and a policy representation in a third. The framework view is therefore necessary to preserve both technical commonality and domain-specific constraints. For evaluation, the treemap suggests that metrics should be selected by role rather than only by model category. Broad representation and generation works can reasonably report reconstruction loss, FVD, perceptual quality, representation linear-probe accuracy, or likelihood-style objectives. Strict world-model works need stronger evidence because their predictions influence decisions. They should report closed-loop return, planning improvement, action feasibility, safety violation rate, calibration, uncertainty-aware abstention, and counterfactual consistency. A visually large block in the treemap does not automatically imply a mature evaluation culture; it may instead indicate a field where many works exist but few test consequence reliability under intervention. The strict/broad layer is especially useful for identifying when claims about world modeling are overextended. A broad video model can be an important world-model substrate, yet it should not be evaluated as if it were already a safe simulator for robotics or driving. A broad LLM tool emulator may be useful for scenario generation, yet it must be checked against executable state before being trusted for irreversible actions. A broad scientific surrogate may learn accurate short-term dynamics, yet it becomes agent-facing only when connected to planning, control, or verification. The role layer therefore acts as a guardrail against conflating predictive capacity with operational reliability. The treemap further clarifies where benchmark design is most urgent. Large broad regions need bridges to action-conditioned evaluation: visual models need controllability tests, representation models need downstream planning tests, and language-based emulators need execution-grounded side-effect checks. Large strict regions need stress tests for distribution shift: model-based reinforcement learning needs model-exploitation diagnostics, driving needs rare-event closed-loop simulation, robotics needs contact and recovery evaluation, and digital agents need hidden-state and permission-state checks. The figure therefore points not only to what has been studied, but also to which evaluation gaps are structurally exposed by the corpus. For future dataset construction, the hierarchy suggests that labels should record more than domain and year. A useful world-model benchmark should annotate state representation, action interface, conditioning signal, transition target, uncertainty signal, simulator or verifier availability, and whether the task is strict or broad. Without these tags, corpus counts become hard to interpret: two papers may both be called world models while one predicts masked embeddings from video and the other simulates browser state after a destructive API call. The treemap makes this heterogeneity visible and motivates richer metadata for future surveys. For system design, the dominant blocks indicate that practical world models are likely to be hybrid. LLM/VLM stacks provide semantic grounding, diffusion and video stacks provide multimodal future generation, latent dynamics provides efficient rollout, code and simulators provide exact or semi-exact checks, and 3D/BEV models provide structured spatial state. The important design question is not which single stack will win, but how these stacks should be composed so that learned prediction remains auditable and actionable. The treemap makes this compositional pressure visible because the largest strict domains already combine several stacks rather than relying on one representation. For reader navigation, the figure offers a compact index into the rest of the paper. A reader interested in embodied control can begin with the robotics and model-based RL blocks, then follow the VLA/WAM section for action coupling. A reader interested in autonomous systems can compare the driving block with the failure-mode and hybrid-verification sections. A reader interested in generative video can start from the vision/video block, then examine why visual realism alone is insufficient for planning. A reader interested in verification can use the auxiliary-verification block to trace hidden state, execution feedback, and tool safety. The main limitation of this treemap is that it encodes paper count, not epistemic certainty. Counts are shaped by publication volume, keyword visibility, corpus selection, and the heuristic classifier used to assign entries to domain, stack, and role. Some small blocks contain influential systems, while some large blocks include broad foundations that are only indirectly agent-facing. The treemap is therefore best understood as an overview of literature density and technical adjacency. It complements, but does not replace, the detailed analysis of formal objectives, architectures, applications, and failure modes in the main survey.

A.4 Practical Reading Guidelines

When using the treemap as a reading guide, the first useful question is whether the work under inspection is strict or broad. Strict papers should be read for their action interface, state transition target, planning mechanism, and evidence that prediction improves behavior. Broad papers should be read for transferable mechanisms: tokenization, representation learning, generative rollout, synthetic data construction, benchmark design, or neural simulation. This separation keeps the survey from making an overly strong claim about works that are technically relevant but not yet closed-loop world models. The second useful question is which state representation the paper treats as authoritative. In model-based reinforcement learning, the authoritative state is often a latent belief optimized for return. In driving, it is often BEV, occupancy, trajectory, or map-conditioned scene state. In auxiliary verification, it may be a mixture of screenshot, accessibility tree, DOM, code state, permission state, and tool trace. In robotics, it may be image, language, proprioception, action history, object state, or a learned latent. The treemap encourages readers to compare these state commitments before comparing model architectures. The third useful question is how uncertainty enters the system. A broad visual model may provide diverse samples without calibrated risk. A strict planner needs uncertainty that can affect action selection, abstention, safety checks, or online correction. A digital-agent emulator needs uncertainty over hidden side effects and execution state. A driving simulator needs uncertainty over rare but safety-critical futures. The domain and role layers therefore indicate where uncertainty is merely a generative property and where it must become an operational control signal. The fourth useful question is what would make a broad block become strict. For visual generation, this usually requires action conditioning, controllability, state persistence, and downstream task evidence. For representation learning, it requires showing that predicted features improve planning or policy learning. For LLM/VLM reasoning, it requires grounding predicted consequences in tool execution, browser state, or verifiable environment updates. For scientific simulation, it requires connecting dynamics prediction to intervention, optimization, or safety-relevant control. These transitions are the natural growth paths suggested by the treemap. The fifth useful question is whether evaluation matches deployment risk. A small digital-agent benchmark involving irreversible tool calls can demand stricter evaluation than a large video-generation benchmark with no action loop. A robotics benchmark with contact-rich manipulation may need physical feasibility checks even if the visual output appears plausible. A driving benchmark must account for closed-loop feedback and rare events. The treemap is useful precisely because it makes this mismatch visible: literature volume, visual appeal, and operational risk do not scale together. Finally, the treemap supports a more disciplined reading of future work. Claims about general world models should specify which parts of the map they cover, which stacks they combine, which domains remain untested, and whether the evidence is strict or broad. A strong future system should not merely occupy a larger rectangle; it should connect rectangles that are currently separate, such as video prediction with action verification, language grounding with executable state, or latent dynamics with calibrated safety checks. In this sense, the figure is both a summary of the current corpus and a compact agenda for integration. A final practical convention is to treat every treemap block as a hypothesis generator. A large block suggests that the field has accumulated enough work for comparative evaluation, but it may also indicate redundancy or metric saturation. A small block suggests either an underexplored opportunity or a domain where data, safety, or system cost makes progress difficult. Reading the map in this way turns corpus visualization into a research-planning tool rather than a decorative summary. This convention is especially important for cross-domain transfer. A stack that appears mature in one domain may fail when transferred to another because observations, action spaces, hidden state, and safety requirements change. For example, video prediction techniques can help robotics, but contact-rich manipulation requires feasibility checks; LLM/VLM reasoning can help auxiliary verification, but tool execution needs exact state; BEV prediction can help driving, but closed-loop control requires rare-event robustness. The treemap therefore encourages transfer with explicit caveats rather than simple architectural analogy.

A.5 Compact Citation Coverage

To preserve citation auditability without the long classification table, the complete corpus is cited below in compact domain groups. **general AI** (120): [88, 170, 276, 235, 9, 105, 280, 300, 113, 290, 3, 256, 265, 165, 71, 249, 43, 117] [50, 96, 116, 21, 83, 17, 12, 79, 7, 174, 301, 255, 77, 257, 134, 187, 219, 216] [188, 241, 269, 230, 142, 401, 242, 126, 38, 308, 326, 356, 329, 146, 55, 106, 260, 49] [25, 24, 346, 232, 1, 119, 302, 405, 40, 190, 365, 395, 153, 328, 222, 182, 47, 131] [183, 271, 4, 402, 343, 200, 75, 399, 5, 212, 203, 161, 91, 345, 294, 18, 295, 287] [29, 163, 236, 157, 90, 250, 171, 319, 275, 168, 103, 70, 175, 220, 125, 193, 380, 196] [266, 53, 336, 128, 312, 8, 213, 205, 233, 39, 41, 362]. **robotics** (69): [74, 148, 33, 63, 154, 292, 56, 385, 73, 2, 278, 145, 364, 273, 274, 137, 30, 180] [228, 136, 223, 195, 66, 305, 221, 316, 363, 375, 331, 352, 286, 172, 396, 307, 178, 246] [392, 379, 45, 202, 80, 42, 400, 267, 252, 387, 94, 209, 388, 150, 217, 330, 207, 397] [344, 342, 54, 282, 378, 173, 86, 204, 291, 371, 369, 85, 52, 152, 293]. **vision/video** (56): [87, 208, 304, 149, 15, 69, 167, 298, 59, 337, 93, 121, 114, 122, 277, 303, 123, 99] [391, 28, 314, 102, 160, 22, 34, 76, 310, 360, 98, 296, 23, 11, 26, 19, 129, 323] [361, 263, 372, 198, 115, 384, 101, 144, 325, 118, 404, 313, 281, 354, 334, 6, 403, 140] [133, 14]. **auxiliary verification** (43): [285, 321, 32, 181, 270, 197, 347, 68, 398, 159, 333, 253, 229, 349, 261, 239, 245, 335] [288, 176, 258, 201, 338, 386, 124, 16, 318, 348, 351, 279, 225, 268, 51, 84, 192, 317] [366, 367, 214, 82, 57, 191, 179]. **autonomous driving** (36): [120, 89, 72, 37, 44, 324, 357, 62, 20, 64, 381, 185, 132, 206, 130, 141, 377, 322] [243, 46, 10, 127, 315, 393, 368, 389, 340, 151, 390, 254, 247, 311, 224, 283, 358, 184]. **interactive worlds** (36): [234, 353, 35, 299, 155, 341, 78, 339, 237, 61, 162, 104, 81, 13, 218, 199, 382, 374] [383, 327, 332, 156, 284, 350, 211, 306, 355, 97, 264, 297, 177, 373, 143, 100, 376, 289]. **model-based RL** (32): [226, 67, 320, 248, 58, 227, 36, 108, 147, 138, 166, 107, 110, 111, 262, 112, 370, 210] [60, 109, 48, 139, 189, 359, 259, 92, 272, 194, 244, 169, 309, 135]. **scientific simulation** (13): [240, 186, 95, 251, 238, 164, 27, 231, 158, 31, 394, 215, 65].

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